

Assignment Specifications

Aims	• Enable students to analyse and employ appropriate Natural Language Processing techniques to solve problems and design intelligent systems. • Enable students to use relevant tools and technology, such as Python programming, to develop intelligent computer programs.
Learning Outcomes Assessed	• Discuss the concepts and importance of natural language in real-world scenarios. (C2, PLO1) • Practice natural language processing algorithms by using the features and capabilities of natural language processing software. (P3, PLO3)
Tutor	Ms Nik Nur Adlin Binti Nik Qausbee Dr Chong Jin Hui
Outline of Problem	For your assignment, you should critically evaluate current technologies, and then propose a project for the selected Natural Language Processing topic, implement an NLP solution to solve the problem in the proposed project using Python/any suitable tools/Framework.
Details	• The Assignment consists of TWO (2) parts. Refer to Project Details Part 1 and Project Details Part 2. Both parts are related. For Part 2, you are required to develop the program using Python. • Group yourself into a team of TWO OR THREE (2-3) members. You are given FOUR (4) options of research areas as listed below. Your group shall select one of them for this assignment. • Please note that the following are the basic requirements of the assignment. Fulfilling the requirements may lead you to an Average or Good grade. Achieving an Excellent grade requires evident extra efforts, such as learning new skills, introducing new ideas, complex NLP algorithms, demonstrating the ability to process big data, and/or producing excellent reports with a working prototype. • Your group is expected to produce ideas that originated from the members, but not to take the work or an idea of someone else (including the Web) and pass it off as your own. NO GROUP IS ALLOWED to share the same idea, i.e., each group must propose a unique title or solution.
Title	1. Open Pre-trained Transformers (OPT) a. Identify a use case that can be applied using Open Pre-trained Transformers (OPT), a suite of decoder-only and, well, pre-trained transformers, ranging from 125 million to 175 billion parameters from Meta.

	b. Perform a background study on the problem and the method to be used. c. Perform data preprocessing and data representation (if any). d. Propose your unique solution using the OPT open source library. e. Hint: students may refer to the webpage for idea and the dataset • https://slator.com/meta-open-sources-its-latest-large-language-model/ • https://github.com/facebookresearch/metaseq/tree/main/projects/OPT
	2. Document Management System (DMS using Named Entity Recognition) a. Identify a problem related to the Document Management System b. Perform a background study on the problem and the method to be used. • You are allowed to determine what is the best document/type is for your project (Resume, Business Document, Receipt, Image) c. Functionality of your DMS: • Perform Entity Named Entity Recognition on the documents (OCR might be involved) • Allow the user to search the entity from the database. • Return the documents with the highlighted entity. • If the word/query strings are not found in the document, return also a similar entity (System will return all the documents with "Kuala Lumpur" if "Johor Bahru" is not found inside the database. d. Hint: Students may refer to the webpage for the idea and the dataset • https://malaya.readthedocs.io/en/latest/load-entities.html
	3. Lemmatiser for Malay/Malaysian Rojak Language a. Perform a background study of the problem/scenario and why it is important b. Research about the Malay language, Malaysia Rojak Language, Malaysia Informal Language, out-of-vocabulary words used in Malaysia. c. General key activities including but not limited to: • Crawl your own data from social media (determine the best platform). • Perform lemmatisation on the data • Allow the user to enter words and determine the lemmatisation word of the word entered d. Reference: Students may refer to the webpage for the idea and the dataset

	• https://malaya.readthedocs.io/en/latest/load-stemmer.html
	4. Rojak/Mixed Language Chatbot - Identify a real-life scenario in which a chatbot can be introduced to solve the problem of spending or procurement activity. For example: handle Q&A to improve customer service experience. - Perform a background study of the problem/scenario and the chatbot to be implemented, with the expected functionalities. - Your designed chatbot should be able to: - Convert Speech to Text and converse with a chatbot. - Learn slang and tone of voice. - Allow humans to correct it and learn through the time. Answer the question using a knowledge-based approach - Design system testing routine for chatbot. For example: • Search for an online dataset. Test your chatbot to evaluate how well their responses are. • Design a real-life testing environment, by asking your classmates to test the chatbot. • The tutor can test the chatbot bypass in some sentences and a logical output is able to be generated by the chatbot, for 2-3 minutes. • Compare and discuss the outcome of the chatbot, and how to improve the chatbot's responses. Reference: Students may refer to the webpage for the idea and the dataset: https://www.pandorabots.com/ https://apps.worldwritable.com/tutorials/chatbot https://www.ibm.com/watson/how-to-build-a-chatbot/
Submission Deadlines	Submit Prototype Source Code and Documentation by: • Source Code: Week 12, Monday, 31st August 2026 before 11.59pm • Documentation: Week 12, Monday, 31st August 2026, before 11.59pm <i>Please submit to the Google Classroom. Late submission will be penalised. A demo session to present the prototype is required.</i>
Contribution	Assignment Part 1: 40% Assignment Part 2: 60% <i>Please submit to the designated Google Classroom.</i>

Academic Integrity and Plagiarism	There must be originality in your work, i.e., do not copy or refer to other group(s). You may only work with your team member(s) to produce the solution of this assignment. You must not share with or refer to any part of the assignment (including the code) of anyone else except your team member(s) and your tutor. Before submitting your assignment, please make sure that you have complied with the TARUMT plagiarism Policy. Any attempt to cheat, plagiarism, collusion, or any other attempt to gain an unfair advantage in the assessment will cause the students concerned to be penalised. IMPORTANT: Students must ensure that the dataset used is obtained legally and ethically. Students are responsible for respecting copyright, privacy, and intellectual property rights. Any AI-generated content or tools used (e.g., ChatGPT, Gemini, Copilot) must be declared appropriately, and students must ensure the originality of their work. Any form of plagiarism or misuse of AI tools will be subject to the TAR UMT plagiarism policy .
Late Submission	Late submission without a valid reason will NOT be tolerated. For late submission, there will be a reduction of total marks: • Late 1 to 3 days after the deadline of submission: minus 10 marks • Late 4 to 7 days after the deadline of submission: minus 20 marks • Late more than 7 days after the deadline of submission: 0 marks In certain circumstances, a student may be allowed to submit the assignment late with a valid reason. S/he must contact the tutor at least one week before the assignment is due. The tutor will evaluate whether the circumstance warrants submitting the assignment late, but no guarantee that the students will not be penalized. Failure to submit the reports & code will result in failure of the coursework.

Project Details: Part 1 Proposal Writing

Introduction	Your task for this part of the assignment is to identify a problem for the selected NLP topic, perform a literature review and propose your respective NLP solution(s) that help in solving the problem in the proposed project.
What to hand in?	The report shall be completed using the given Google Doc template in the Google Classroom
Format for Deliverable	For your proposal, you are required to complete the report using the Google Doc template available in the Google Classroom. Each group leader must turn in the individual report to the Google Classroom before the stipulated deadline. Otherwise the late submission penalty will be applied.

Project Details: Part 2 Prototype Development

Introduction	Your task for this assignment is to implement the NLP solution using Python or any other relevant tools, perform testing and evaluation on the system and finally present the work to your tutor.
What to hand in?	Submit all your source code to Google Classroom submission page. Each team member is required to do a presentation of the work and Q&A session in Week 13 or Week 14 based on the arrangement with the tutor.
Format for Deliverable	Compress the entire source code using zip format. Submit the source code for each group leader.